Atomic physics – the structure of atoms and their spectra

Every element has its own fingerprint of spectral lines. From them physicists have determined what the stars are made of, discovered helium on the Sun before it was found on Earth, and found the laws of quantum mechanics. Atomic physics explains why atoms have definite energies, why the lines split in magnetic fields and how the periodic table is built up – and today it is the basis of atomic clocks, laser cooling and quantum computers. This page covers the subject with three interactive figures.

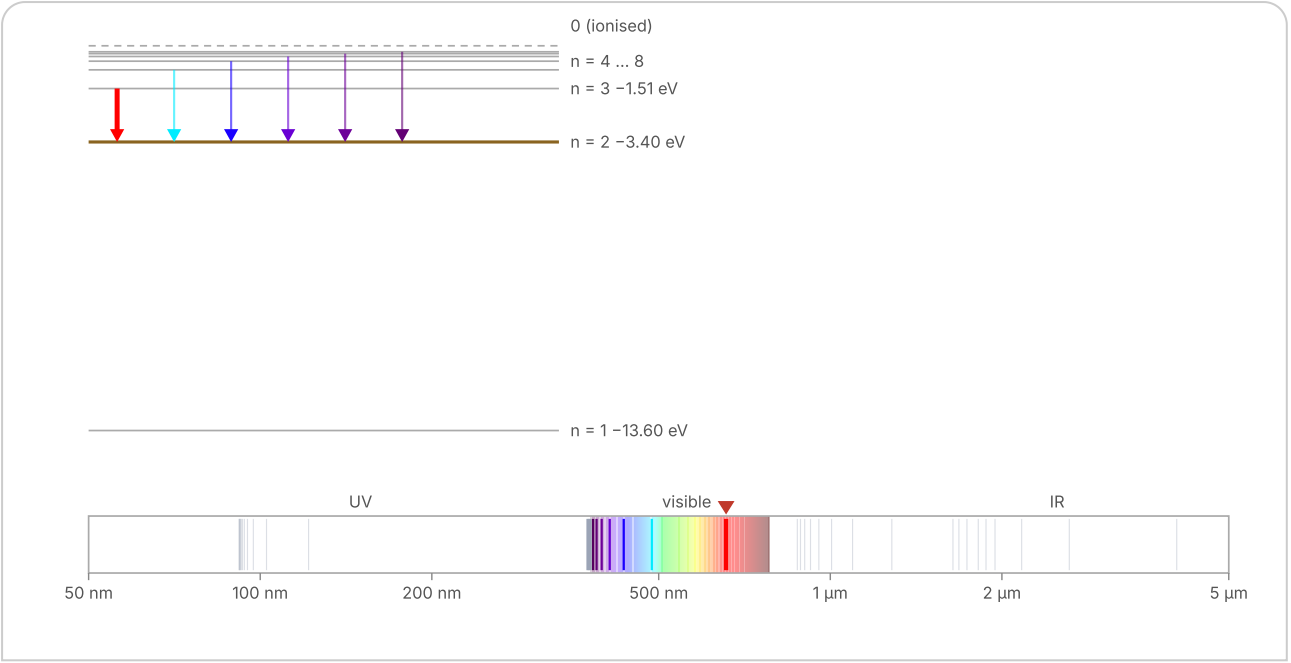
Spectral lines – the fingerprints of atoms

In 1814 Joseph von Fraunhofer saw hundreds of dark lines in the solar spectrum. In Heidelberg in 1859–60 Gustav Kirchhoff and Robert Bunsen showed that each element emits and absorbs light at quite definite wavelengths. Sodium gives the yellow doublet at 589 nm, and the dark D lines in the Sun's spectrum are due to sodium in the solar atmosphere. With the spectroscope Bunsen and Kirchhoff discovered caesium and rubidium, and in 1868 Jules Janssen and Norman Lockyer saw a yellow line in the Sun's corona that matched no known substance: helium, first found on Earth in 1895.

Hydrogen, the simplest atom, has the simplest spectrum. In 1885 the Swiss schoolteacher Johann Balmer found a formula for its four visible lines, and in 1888 Johannes Rydberg generalised it:

$$\frac{1}{\lambda} = R_\infty Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right), \qquad R_\infty = 1.097 \cdot 10^7 \ \mathrm{m}^{-1}$$

With $n_1 = 1$ one gets the Lyman series in the ultraviolet (1906), with $n_1 = 2$ the Balmer series in the visible and with $n_1 = 3$ the Paschen series in the infrared (1908). The Rydberg constant is today one of the best-known constants of nature, to 12 significant figures.



Bohr’s model of the atom (1913)

In 1911 Rutherford had shown that the atom has a small, heavy nucleus (see Nuclear physics). But according to Maxwell’s theory an electron orbiting a nucleus should radiate and fall into the nucleus within a hundred-millionth of a second. Niels Bohr, aged 27 and working with Rutherford in Manchester, postulated in 1913 that the electron can only occupy certain stationary orbits with angular momentum $L = n\hbar$, and that it emits a photon when it jumps from one orbit to another:

$$E_n = -\frac{m_e e^4}{8\varepsilon_0^2 h^2}\,\frac{Z^2}{n^2} = -13.6\ \text{eV}\,\frac{Z^2}{n^2}, \qquad r_n = a_0\frac{n^2}{Z}, \quad a_0 = 0.0529\ \text{nm}, \qquad h\nu =$$

The model gave Rydberg’s formula and the value of the Rydberg constant from e, m_e and h – a triumph. It also explained the lines from ionised helium that Pickering had attributed to hydrogen. In 1914 James Franck and Gustav Hertz showed that atoms take up energy only in definite portions. But the Bohr model could not explain atoms with several electrons, the intensities of the lines, or why the orbits were stable. The answer came with quantum mechanics in 1925–26.

The hydrogen atom in quantum mechanics

Schrödinger’s equation for an electron in the Coulomb field of a nucleus can be solved exactly (see Quantum mechanics). The solutions are labelled by three quantum numbers:

Principal quantum number n

$n = 1, 2, 3, \ldots$ determines the energy $E_n = -13.6\ \text{eV}/n^2$ and the size of the orbital.

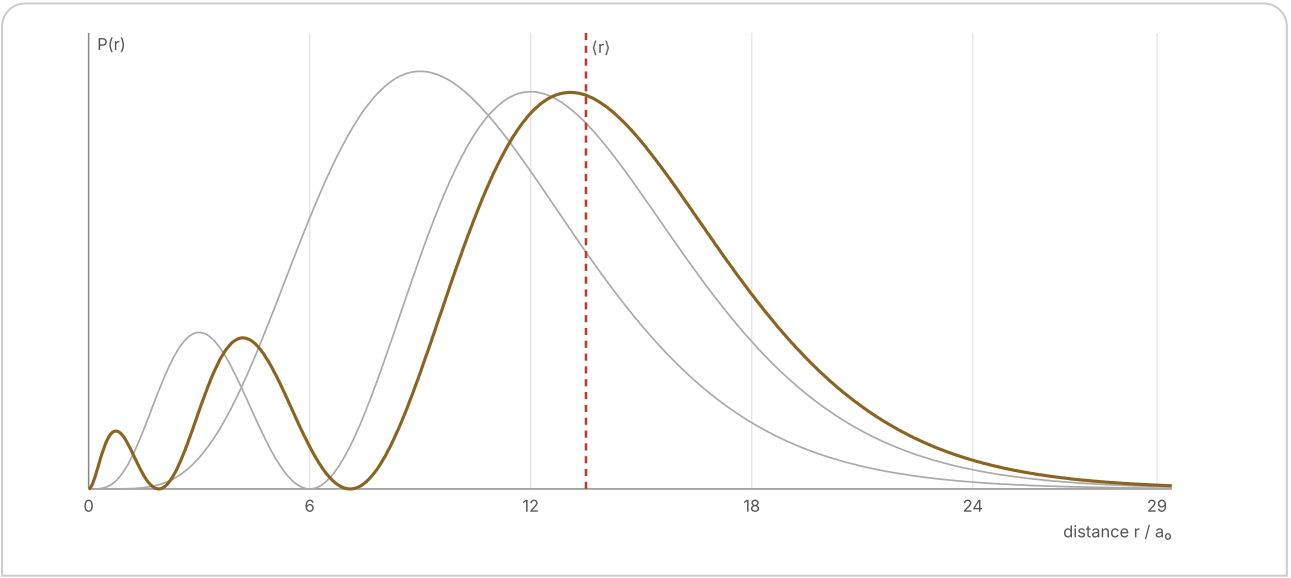
Orbital quantum number l

$l = 0, 1, \ldots, n-1$ (labelled s, p, d, f) determines the angular momentum $|\vec{L}| = \sqrt{l(l+1)}\,\hbar$ and the shape of the orbital.

Magnetic quantum number m_l

$m_l = -l, \ldots, l$ determines the component of angular momentum along an axis, $L_z = m_l \hbar$.

The wave function is a product of a radial part and a spherical harmonic, $\psi_{nlm} = R_{nl}(r)\,Y_{lm}(\theta, \varphi)$. The probability of finding the electron at a distance between r and $r + dr$ from the nucleus is $P(r)\,dr = r^2 |R_{nl}|^2\,dr$. For the ground state it peaks exactly at the Bohr radius a_0 – but the electron has no orbit. See the interactive orbitals.



Fine structure and the electron’s spin

With a good spectroscope Hα is not one line but several closely spaced ones. The electron has an intrinsic angular momentum, *spin* $s = \frac{1}{2}$ (Uhlenbeck and Goudsmit, 1925), and a magnetic moment. In the electron’s rest frame the nucleus orbits, which produces a magnetic field with

which the spin interacts: *spin–orbit coupling*. The orbital and spin angular momenta add to $\vec{J} = \vec{L} + \vec{S}$ with $j = l \pm \frac{1}{2}$. Together with the relativistic corrections this gives the fine structure

$$\Delta E_{\text{fine}} \sim \alpha^2 \, |E_n|, \qquad \alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c} \approx \frac{1}{137.036}$$

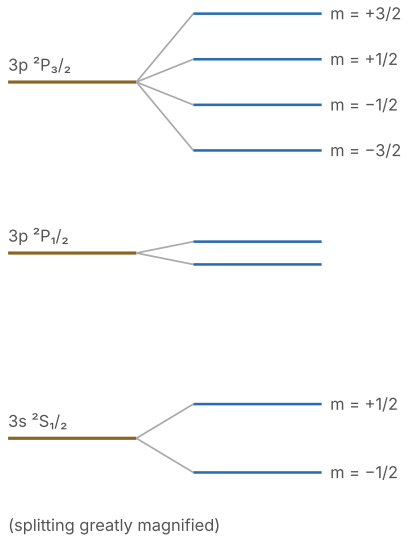
The fine-structure constant $\boldsymbol{\alpha}$, introduced by Sommerfeld in 1916, is one of the pure numbers of physics. Dirac's relativistic equation (1928) gave the fine structure exactly. In sodium the spin–orbit coupling gives the famous yellow doublet: the 3p level splits into $3p_{1/2}$ and $3p_{3/2}$, and the D lines lie at 589.6 and 589.0 nm. In 1947 Willis Lamb and Robert Retherford measured that $2s_{1/2}$ and $2p_{1/2}$ in hydrogen – which Dirac said should have the same energy – are separated by 1057 MHz. The Lamb shift is due to vacuum fluctuations and was explained by quantum electrodynamics (Feynman, Schwinger, Tomonaga).

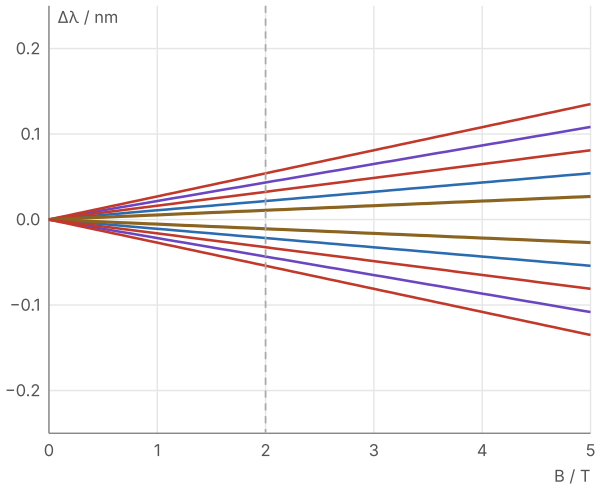
Atomic states are written in spectroscopic notation $^{2S+1}L_J$, e.g. $^2S_{1/2}$ for the ground state of hydrogen and sodium and $^2P_{3/2}$ for the upper D level of sodium.

The Zeeman effect

In 1896 in Leiden Pieter Zeeman placed a sodium flame between the poles of an electromagnet and saw the D lines broaden. Hendrik Lorentz explained this with electrons oscillating in the atom and predicted that the lines should split into three and be polarised – this was confirmed, and the two shared the Nobel Prize in 1902 (see Lorentz). But many lines split into more components – the "anomalous" Zeeman effect, which was only understood with spin. A level with angular momentum J splits into $2J+1$ sublevels:

$$\Delta E = g_J \, m_J \, \mu_B B, \qquad \mu_B = \frac{e\hbar}{2m_e} = 57.9\ \mu\text{eV/T}, \qquad g_J = 1 + \frac{J(J+1) + S(S+1)}{2J(J}$$





Today the Zeeman effect is used to measure magnetic fields in sunspots (George Ellery Hale, 1908) and in distant stars. The corresponding splitting in an electric field is called the Stark effect (1913). In a magnetic field one can also flip the electrons' spins directly with microwaves – electron spin resonance – or the spins of nuclei with radio waves, as in the MRI scanner.

Atoms with many electrons

For all atoms heavier than hydrogen the Schrödinger equation cannot be solved exactly, because the electrons repel each other. The most important approximation is the *central-field model*: each electron moves in an average field from the nucleus and the other electrons. Douglas Hartree (1928) and Vladimir Fock (1930) developed the self-consistent method, in which the field is calculated from the electrons' own wave functions until the two agree. Today density functional theory (Walter Kohn, Nobel Prize in Chemistry 1998) is used throughout chemistry and materials science.

Electrons are fermions, and Wolfgang Pauli's exclusion principle (1925) says that no two electrons can have all four quantum numbers (n, l, m_l, m_s) the same. A shell with principal quantum number n can therefore hold at most $2n^2$ electrons: 2, 8, 18, 32. The electrons in the inner shells screen the nuclear charge, so the outer electrons feel an *effective* charge Z_{eff}. Orbitals with small l penetrate further towards the nucleus and are bound more strongly – which is why 4s lies below 3d. Friedrich Hund's rules (1925) determine how the electrons in a partly filled shell arrange themselves: maximum total spin first. That is why oxygen is paramagnetic and iron is magnetic.

Helium is the simplest example. Its two electrons can have parallel spins (orthohelium, triplet) or opposite spins (parahelium, singlet), and the two forms have such different spectra that for a time they were thought to be two different substances. Heisenberg explained this in 1926 with the *exchange interaction* – a purely quantum-mechanical effect that also lies behind ferromagnetism.

The periodic table

In 1869 Dmitri Mendeleev arranged the 63 known elements by atomic weight and chemical properties, leaving gaps for elements not yet discovered. He predicted gallium, scandium and germanium together with their properties – they were found in 1875–86. But why is the table periodic? The answer came from atomic physics: in 1921–22 Bohr explained the structure of the periodic table with shells, and predicted that element 72 must resemble zirconium and not the rare earths. Dirk Coster and George de Hevesy found it in 1923 at Bohr's institute and named it hafnium after the Latin name of Copenhagen.

The periods have the lengths 2, 8, 8, 18, 18, 32, 32, corresponding to the filling order 1s, 2s 2p, 3s 3p, 4s 3d 4p, 5s 4d 5p, 6s 4f 5d 6p, … (the Madelung rule). The ionisation energy is greatest for the noble gases with filled shells and smallest for the alkali metals with one electron outside a filled shell. See The periodic table.

X-ray spectra and Moseley

In 1913 in Manchester Henry Moseley measured the wavelength of the characteristic X-rays from 38 elements. When an electron is knocked out of the innermost K shell, an electron from the L shell drops down and emits the Kα radiation. Moseley found that the square root of the frequency grows linearly with a whole number, the atomic number:

$$\sqrt{\nu_{K\alpha}} \propto Z - 1 \qquad \Leftrightarrow \qquad h\nu_{K\alpha} \approx 13.6\ \text{eV}\,(Z-1)^2\left(1 - \frac{1}{4}\right)$$

The one electron remaining in the K shell screens the nuclear charge by one unit – hence $Z - 1$. Moseley's law showed that the order of the elements is determined by the nuclear charge, not the atomic weight, and it revealed the gaps that remained: elements 43, 61, 72 and 75. Moseley was killed at Gallipoli in 1915, aged 27.

Transitions, lifetimes and selection rules

An excited atom decays spontaneously with a lifetime that is typically nanoseconds for allowed transitions. The light comes from an oscillating electric dipole moment, and this gives *selection rules*: $\Delta l = \pm 1$, $\Delta m_l = 0, \pm 1$, and the spin does not change. The transition from 2s to 1s in hydrogen is "forbidden" and takes 0.12 s via two photons. Forbidden lines can be seen in thin nebulae in space, where the atoms do not collide – the green lines from doubly ionised oxygen, [O III], were attributed in the 1860s to a new element, "nebulium", until Ira Bowen explained them in 1927.

The lines have a natural width $\Delta\nu = 1/(2\pi\tau)$ from the lifetime, but in a gas they are usually broader because of the Doppler effect from the motion of the atoms and pressure broadening from collisions. The intensity, width and shift of the lines give astronomers the temperature, pressure, composition and velocity of the stars. See The interaction of light with matter.

Hyperfine structure and atomic clocks

The nucleus also has spin and a small magnetic moment. Its interaction with the electron gives the *hyperfine structure*, about a thousand times smaller than the fine structure. In the ground state of hydrogen there are two levels: electron and proton spins parallel or opposite. The transition between them gives the radio line at 1420.4 MHz – the 21 cm line – predicted by Henk van de Hulst in 1944 and measured by Ewen and Purcell in 1951. It has mapped the spiral arms of the Milky Way.

In caesium-133 the hyperfine transition of the ground state is 9 192 631 770 Hz, and since 1967 the second has been defined as that many oscillations. In the 1930s Isidor Rabi developed the molecular-beam resonance method, Norman Ramsey the method of separated fields, and the first caesium clocks were built in 1955. Modern optical atomic clocks with trapped ions or atoms in optical lattices are so precise that they can measure the time difference when the clock is raised by one centimetre in the Earth's gravitational field (see Relativity), and a new definition of the second is being prepared.

Modern atomic physics

Laser cooling

Atoms can be slowed by laser light striking them head-on, to temperatures of microkelvins (Chu, Cohen-Tannoudji and Phillips, Nobel Prize 1997). This led to Bose–Einstein condensates in 1995 (Nobel Prize 2001). See Laser physics.

Single atoms and ions

Hans Dehmelt and Wolfgang Paul trapped single ions in electromagnetic traps (Nobel Prize 1989); Serge Haroche and David Wineland controlled individual quantum systems (2012). Trapped ions are today one of the leading types of quantum computer.

Rydberg atoms

Atoms excited to $n \sim 100$ are almost a micrometre across and interact strongly with each other. Arrays of Rydberg atoms in optical tweezers are used as quantum simulators.

Precision tests

Antihydrogen is produced and measured at CERN (ALPHA). If the 1s–2s transition in antihydrogen differed from that in hydrogen, CPT symmetry would be broken; they have been measured to agree to 12 digits.

Further reading

- Christopher J. Foot: *Atomic Physics* (Oxford 2005)
- H. Haken and H.C. Wolf: *The Physics of Atoms and Quanta* (7th ed., Springer 2005)
- B.H. Bransden and C.J. Joachain: *Physics of Atoms and Molecules* (2nd ed., Pearson 2003)
- Robert Eisberg and Robert Resnick: *Quantum Physics of Atoms, Molecules, Solids, Nuclei and Particles* (2nd ed., Wiley 1985)
- The biographies of Bohr, Pauli and Schrödinger

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